

KAIJIAN WANG

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EDUCATION

Rice University, Texas, United States 2025 – Present

Ph.D Student in Computer Science, *Advisor:* Prof. Yuke Wang

University of Science and Technology of China, Anhui, China 2021 – 2025

B.E. in Information Security

Courses: Data Structure and Algorithm, Operating System, Compiler Principle, Machine Learning, Network Security Protocol, Computer Network etc.

EXPERIENCES

Research Scientist Intern, Meta Superintelligence Labs - Infra May. 2026 – Sep. 2026

PyTorch Distributed Team *Advisor:* Sanket Purandare

SELECTED PROJECTS

Distributed Video Diffusion Inference on TPU Meshes Aug. 2025 – Now

Research Assistant at Rice *Advisors:* Prof. Yuke Wang, Dr. Yarong Mu

- Designing an auto-parallel strategy for Diffusion Transformer inference on TPU, extending standard TP/DP with sequence parallelism enabled by stale KV reuse across timesteps.
- Building a cost model for TPU mesh topology where ICI bandwidth is path-dependent: communication between distant chips consumes intermediate links, unlike homogeneous GPU NVLink collectives.
- On-going, collaborating with Google TPU team

Efficient Tool-Augmented LLM Serving System Feb. 2024 – Dec. 2024

Research Intern at UC San Diego *Advisors:* Dr. Zheng Wang, Prof. Yufei Ding

- Built a tool-augmented LLM serving system over vLLM with predictive KV cache management: proactive preemption to DRAM on API call, prefetching before return based on predicted latency, and speculative parallel API invocation via a small prediction model.
- Designed dynamic runtime memory reallocation between models—freeing the small model under high load to increase large model batch size, reallocating when load drops.
- Rewrote KV cache scheduling at block granularity to hide transfer latency behind computation; optimized vLLM's CPU-bound swap kernel (2x speedup).
- Implemented nearly the entire framework. Achieved 30-40% end-to-end speedup over large-model-only baseline.

Efficient DLRM Training System over CXL Aug. 2024 – Nov. 2024

Research Intern at UC San Diego *Advisors:* Dr. Zheng Wang, Prof. Yufei Ding

- Designed a hierarchical CXL memory architecture for DLRM training, tiering embedding placement by access frequency across dedicated and shared memory device units (MDUs).
- Formulated joint MDU allocation and embedding placement as MILP with bandwidth and capacity constraints; achieved 1.37x improvement over naive placement.
- Full system achieved 5.34x reduction in end-to-end training step latency.

PUBLICATIONS

- **ALTO:** Adaptive LoRA Tuning and Orchestration for Heterogeneous LoRA Training Workloads [arXiv:2604.05426]

Jingwei Zuo, Xinze Feng, Zien Liu, **Kaijian Wang**, Fanjiang Ye, Ye Cao, Zhuang Wang, Yuke Wang

- **SuperGen:** An Efficient Ultra-high-resolution Video Generation System with Sketching and Tiling
[arXiv:2508.17756]
*Fanjiang Ye, Zepeng Zhao, Yi Mu, Jucheng Shen, Renjie Li, **Kaijian Wang**, Desen Sun, Saurabh Agarwal, Myungjin Lee, Triston Cao, Aditya Akella, Arvind Krishnamurthy, T.S. Eugene Ng, Zhengzhong Tu, Yuke Wang*
- **RecDM:** Efficient Training System for Large-Scale Recommendation Models on Disaggregated Memory
(In submission)
*Zheng Wang, Zhongkai Yu, **Kaijian Wang**, Yichen Lin, Yikai Li, Liu Liu, Xulong Tang, Yuke Wang, Yangwook Kang, Yufei Ding*

SCHOLARSHIPS AND AWARDS

USTC-Fellowship (~7.5k USD), USTC	May. 2025
Outstanding Student Scholarship, USTC	Oct. 2024
DAS Scholarship, USTC & DAS-Security Inc.	Nov. 2023
Outstanding Student Scholarship, USTC	Oct. 2023
Zhang Zongzhi Science and Technology Scholarship, USTC	Oct. 2022
2 nd Prize (top 5 /389), Hackergame2024, USTC	Nov. 2024
Top 3 rd (1%), TPCTF, Tsinghua University & Peking University	Nov. 2023
3 rd Prize (top 10%), Hackergame2023, USTC	Nov. 2023
3 rd Prize in competition area (top 20%), - National College Student Information Security Contest, China	June. 2023
3 rd Prize (top 20%), D^3CTF, Alibaba Inc.	May. 2023

SKILLS

- Programming Language: Python, C/C++, SQL, HTML, $\LaTeX$
- ML related: PyTorch, JAX, vLLM, Recommendation Systems, Diffusion Models
- Research Interest: Efficient LLM Serving/Training, Distributed Systems